Data Structures & Algorithms - Trees, Graphs & Heaps

Course: Data Structures | Semester 3 | Contributor: Alex Chen

AVL rotations, Red-Black balancing, Dijkstra shortest path, and Disjoint Set Union.

Section 1: Balanced Binary Search Trees

* AVL balance factors -1, 0, +1: LL, RR, LR, and RL rotation mechanics.
* Red-Black tree properties: root blackness, red node black children, black-height invariance.
* Binary Heap: array indexing parent/left/right, heapify down/up in Olog N.

Section 2: Graph Algorithms & Minimum Spanning Trees

* BFS for unweighted shortest path, DFS for cycle detection and topological sort.
* Dijkstras algorithm with priority queue in OV + E log V.
* Kruskals MST using Disjoint Set Union with path compression and rank union.

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